Year 11 Higher

REVISION PACK

Shape and Space 6

Name: _____

Class: _____

Shape and Space 6 · Angles in polygons (Lessons 10.2, 10.3, 10.4)

Worked example: regular polygons

(a) The exterior angle of a regular polygon is 40°. Work out the number of sides.

(b) Each interior angle of a regular polygon is 160°. How many sides does it have?

(a)	Step 1	Divide 360 by the exterior angle	$360 \div 40 = 9$
	Step 2	Answer	9 sides
(b)	Step 1	Exterior = 180 − interior	$180 - 160 = 20$
	Step 2	Divide 360 by it	$360 \div 20 = 18$
	Step 3	Answer	18 sides

Practice: regular polygons

1.1 The exterior angle of a regular polygon is 20°. Work out the number of sides of the polygon.

1.2 Each interior angle of a regular polygon is 144°. How many sides does the polygon have?

1.3 The exterior angle of a regular polygon is 36°. Work out the number of sides of the polygon.

1.4 Each interior angle of a regular polygon is 108°. How many sides does the polygon have?

1.5 The exterior angle of a regular polygon is 45°. Work out the number of sides of the polygon.

1.6 Each interior angle of a regular polygon is 135°. How many sides does the polygon have?

Shape and Space 6 · Angles in polygons (Lessons 10.2, 10.3, 10.4)

Worked example: finding x in a polygon

A pentagon has angles 100°, 120°, 95°, x and $2x$. Work out the value of x.

Step 1	Angle sum: (n − 2) × 180	$(5-2)\times 180 = 540$
Step 2	Add the angles you know	$100 + 120 + 95 = 315$
Step 3	Take it from 540°	$3x = 540 - 315 = 225$
Step 4	Divide by 3	$x = 225 \div 3 = 75$
Step 5	Answer	$x = 75$

Practice: finding x in a polygon

1.7

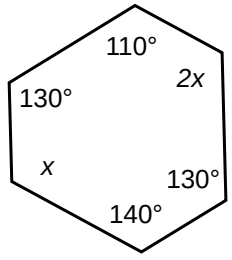


Diagram NOT accurately drawn

The diagram shows a hexagon.

Work out the value of x.

1.8

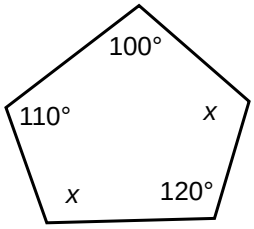


Diagram NOT accurately drawn

The diagram shows a pentagon.

Work out the value of x.

1.9

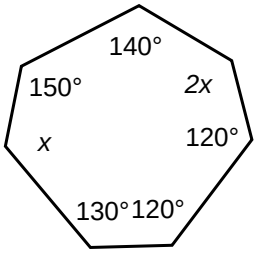


Diagram NOT accurately drawn

The diagram shows a heptagon.

Work out the value of x.

Practice: finding x in a polygon (continued)

1.10

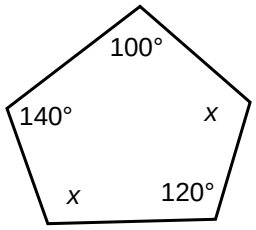


Diagram NOT accurately drawn

The diagram shows a pentagon.
Work out the value of x.

1.11

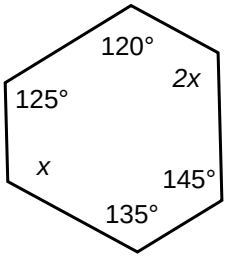


Diagram NOT accurately drawn

The diagram shows a hexagon.
Work out the value of x.

1.12

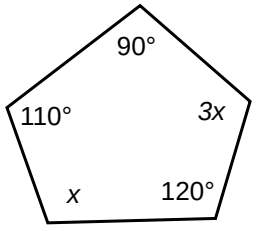


Diagram NOT accurately drawn

The diagram shows a pentagon.
Work out the value of x.

Shape and Space 6 · Angles in polygons (Lessons 10.2, 10.3, 10.4)

Worked example: angles in a kite

$ABCD$ is a kite with $AB = AD$. Angle BAD = 80° and angle BCD = 60°. Work out the size of angle ABC.

Step 1	Angles in a quadrilateral	$360 - 80 - 60 = 220$
Step 2	The other two are equal	$220 \div 2 = 110$
Step 3	Answer	110°

Practice: angles in a kite

1.13

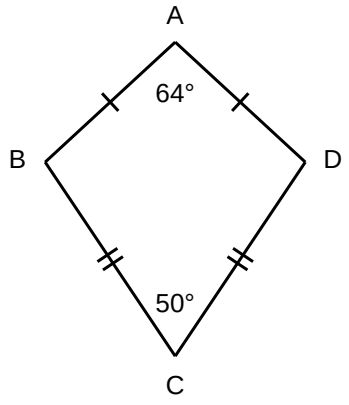


Diagram **NOT** accurately drawn

$ABCD$ is a kite with $AB = AD$.
Angle BAD = 64° and angle BCD = 50°.
Work out the size of angle ABC.

1.14

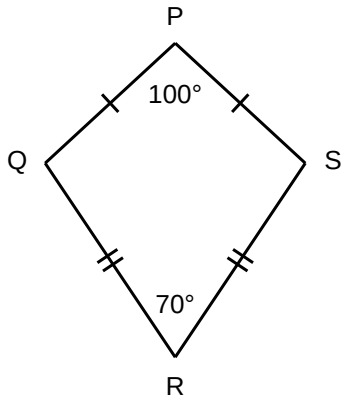


Diagram **NOT** accurately drawn

$PQRS$ is a kite with $PQ = PS$.
Angle QPS = 100° and angle QRS = 70°.
Work out the size of angle PQR.

1.15

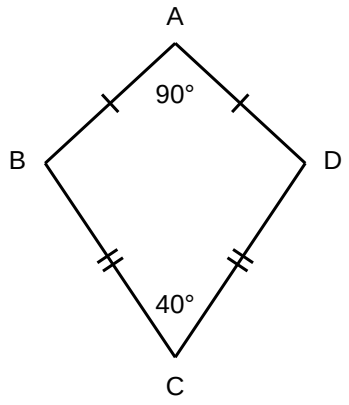


Diagram **NOT** accurately drawn

$ABCD$ is a kite with $AB = AD$.
Angle BAD = 90° and angle BCD = 40°.
Work out the size of angle ABC.

Practice: angles in a kite (continued)

1.16

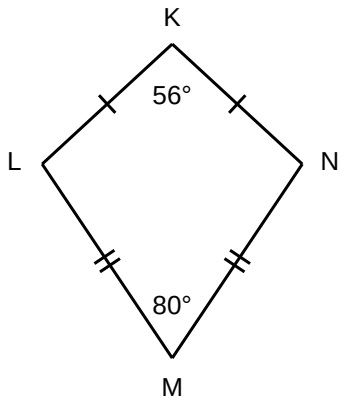


Diagram **NOT** accurately drawn

$KLMN$ is a kite with $KL = KN$.
Angle LKN = 56° and angle LMN = 80°.
Work out the size of angle KLM.

1.17

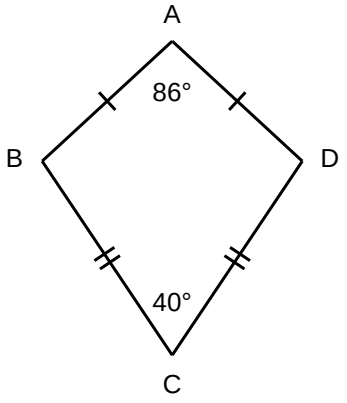


Diagram **NOT** accurately drawn

$ABCD$ is a kite with $AB = AD$.
Angle BAD = 86° and angle BCD = 40°.
Work out the size of angle ABC.

1.18

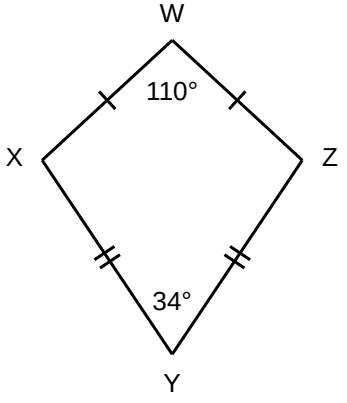


Diagram **NOT** accurately drawn

$WXYZ$ is a kite with $WX = WZ$.
Angle XWZ = 110° and angle XYZ = 34°.
Work out the size of angle WXY.

Shape and Space 6 · Congruence and similarity (Lessons 10.1, 10.6)

Worked example: proving triangles are congruent

$ABCD$ is a square. Prove that triangle ABC is congruent to triangle ADC.

Step 1	Equal sides	$AB = AD$ (sides of a square)
Step 2	Equal sides	$BC = DC$ (sides of a square)
Step 3	Common side	AC is in both triangles
Step 4	Answer	congruent (SSS)

Practice: proving triangles are congruent

2.1

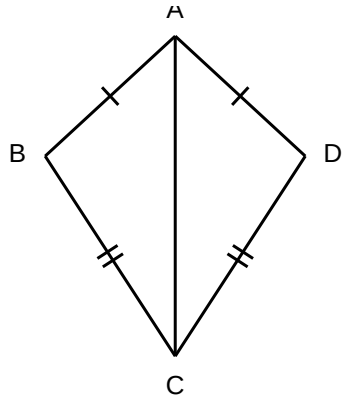


Diagram **NOT** accurately drawn

$ABCD$ is a kite with $AB = AD$ and $CB = CD$.
Prove that triangle ABC is congruent to triangle ADC.

2.2

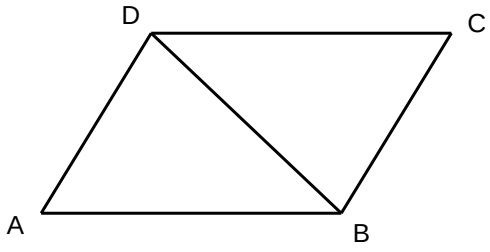


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram. BD is a diagonal.
Prove that triangle ABD is congruent to triangle CDB.

Practice: proving triangles are congruent (continued)

2.3

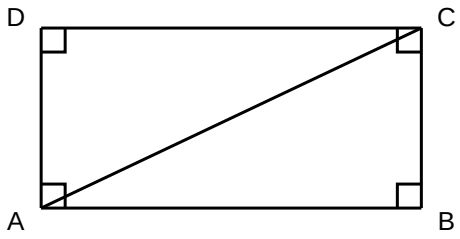


Diagram **NOT** accurately drawn

$ABCD$ is a rectangle. AC is a diagonal.
Prove that triangle ABC is congruent to triangle CDA.

2.4

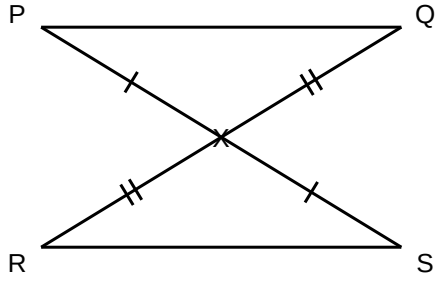


Diagram **NOT** accurately drawn

PS and QR cross at X. $PX = XS$ and $QX = XR$.
Prove that triangle PXQ is congruent to triangle SXR.

Shape and Space 6 · Congruence and similarity (Lessons 10.1, 10.6)

Worked example: similar triangles with parallel lines

DE is parallel to BC. AD = 3 cm, DB = 6 cm and DE = 4 cm.

(a) Work out the length of BC.

(b) AE = 2.5 cm. Work out the length of EC.

(a)	**Step 1**	Whole side	$AB = 3 + 6 = 9$
	Step 2	Scale factor	$9 \div 3 = 3$
	Step 3	Side × scale factor	$BC = 4 \times 3 = 12$
	Step 4	Answer	12 cm
(b)	**Step 1**	Whole side: 2.5 × scale factor	$AC = 2.5 \times 3 = 7.5$
	Step 2	Subtract AE	$7.5 - 2.5 = 5$
	Step 3	Answer	5 cm

Practice: similar triangles with parallel lines

2.5

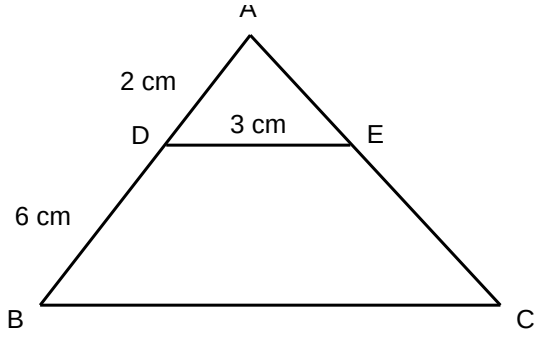


Diagram **NOT** accurately drawn

DE is parallel to BC.

AD = 2 cm, DB = 6 cm and DE = 3 cm.

(a) Work out the length of BC.

(b) AE = 1.5 cm. Work out the length of EC.

2.6

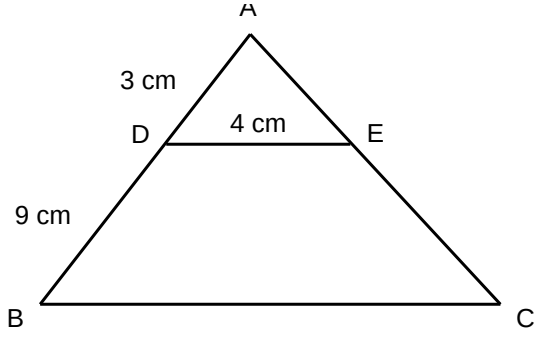


Diagram **NOT** accurately drawn

DE is parallel to BC.

AD = 3 cm, DB = 9 cm and DE = 4 cm.

(a) Work out the length of BC.

(b) AE = 2.5 cm. Work out the length of EC.

Practice: similar triangles with parallel lines (continued)

2.7

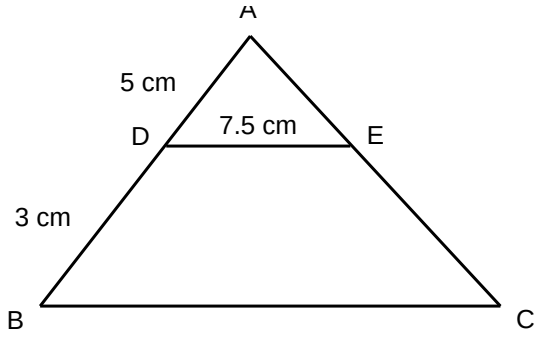


Diagram **NOT** accurately drawn

DE is parallel to BC.

AD = 5 cm, DB = 3 cm and DE = 7.5 cm.

(a) Work out the length of BC.

(b) AE = 4 cm. Work out the length of EC.

2.8

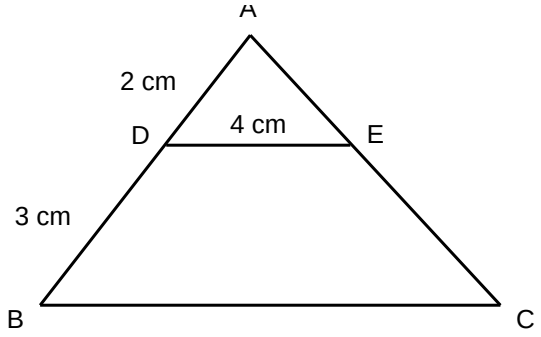


Diagram **NOT** accurately drawn

DE is parallel to BC.

AD = 2 cm, DB = 3 cm and DE = 4 cm.

(a) Work out the length of BC.

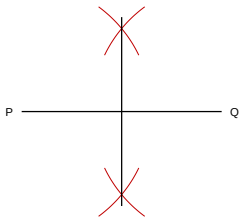
(b) AE = 1.8 cm. Work out the length of EC.

Shape and Space 6 · Congruence and similarity (Lessons 10.1, 10.6)

Worked example: the perpendicular bisector

Draw a line PQ 7 cm long. Construct the perpendicular bisector of PQ.

Step 1	Compasses more than half of PQ
Step 2	Arcs from P, above and below
Step 3	Same arcs from Q
Step 4	Join the two crossings



Practice: the perpendicular bisector

2.9 Draw a line PQ 9 cm long. Construct the perpendicular bisector of PQ.

2.10 Draw a line AB 8 cm long. Construct the perpendicular bisector of AB.

2.11 Draw a line XY 10 cm long. Construct the perpendicular bisector of XY.

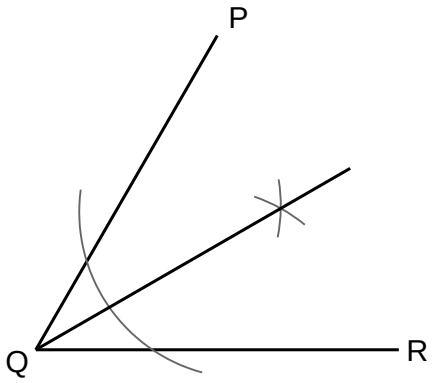
2.12 Draw a line CD 6 cm long. Construct the perpendicular bisector of CD.

2.13 Draw a line EF 11 cm long. Construct the perpendicular bisector of EF.

2.14 Draw a line MN 5 cm long. Construct the perpendicular bisector of MN.

Worked example: the bisector of an angle

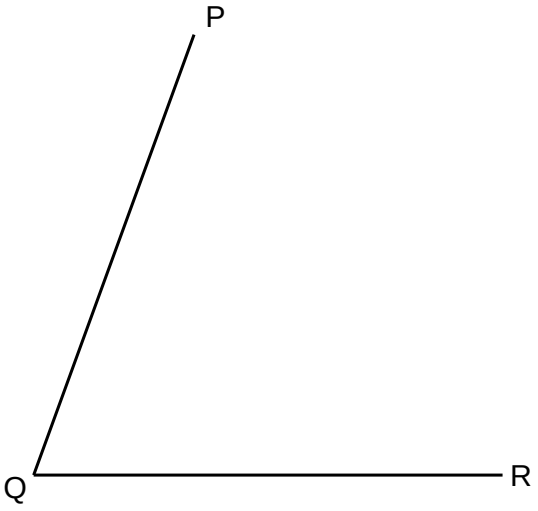
Use ruler and compasses only to construct the bisector of angle PQR.



Step 1	Compasses on Q	draw an arc that crosses both arms
Step 2	Same radius, compasses on each crossing point	draw two arcs that cross inside the angle
Step 3	Join Q to where the arcs cross	this line cuts the angle in half
Step 4	Answer	Leave all the arcs showing

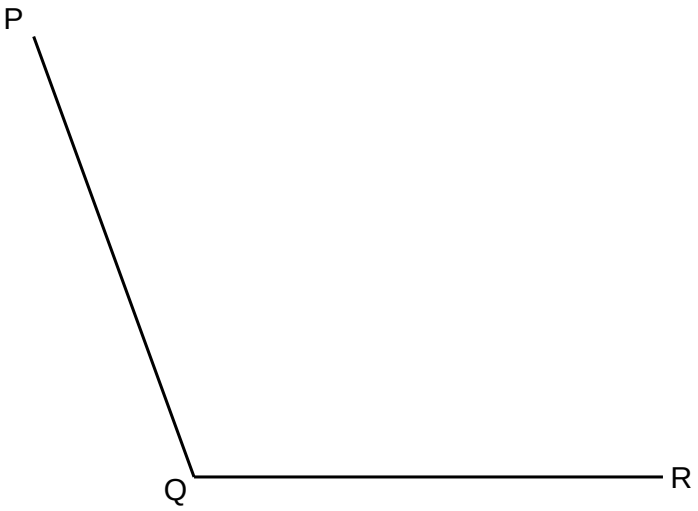
Practice: the bisector of an angle

8.1 Use ruler and compasses only to construct the bisector of angle PQR.
You must show all your construction lines.



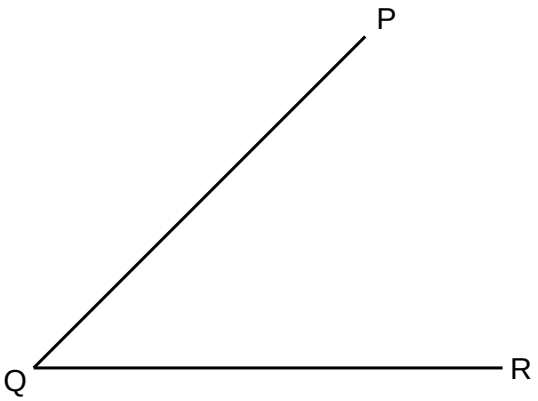
Practice: the bisector of an angle (continued)

8.2 Use ruler and compasses only to construct the bisector of angle PQR.
You must show all your construction lines.



Practice: the bisector of an angle (continued)

8.3 Use ruler and compasses only to construct the bisector of angle PQR.
You must show all your construction lines.



Practice: the bisector of an angle (continued)

8.4 Use ruler and compasses only to construct the bisector of angle PQR.
You must show all your construction lines.

